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CRAFT: Character-Region Awareness for Text Detection

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Character Region Awareness for Text Detection

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Abstract

Scene text detection methods based on neural networks have emerged recently and have shown promising results. Previous methods trained with rigid word-level bounding boxes exhibit limitations in representing the text region in an arbitrary shape. In this paper, we propose a new scene text detection method to effectively detect text area by exploring each character and affinity between characters. To overcome the lack of individual character level annotations, our proposed framework exploits both the given character-level annotations for synthetic images and the estimated character-level ground-truths for real images acquired by the learned interim model. In order to estimate affinity between characters, the network is trained with the newly proposed representation for affinity. Extensive experiments on six benchmarks, including the TotalText and CTW-1500 datasets which contain highly curved texts in natural images, demonstrate that our character-level text detection significantly outperforms the state-of-the-art detectors. According to the results, our proposed method guarantees high flexibility in detecting complicated scene text images, such as arbitrarily-oriented, curved, or deformed texts.

Figure 1. Visualization of character-level detection using CRAFT. (a) Heatmaps predicted by our proposed framework. (b) Detection results for texts of various shape.

most of the existing text datasets do not provide character-level annotations, and the work needed to obtain character-level ground truths is too costly.


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Text detection

axis-aligned/oriented rect
quad, polygon
curved text

word boundaries
ambiguous

link chars
bottom-up

too costly
to annotate

instant translation

image retrieval

document parsing

navigation

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Previous work

- pre deep learning features
  - MSER (Matas et al.), SWT
- bounding box regression
  - adapted from general object detection
  - TextBoxes
- pixel-level segmentation
  - PixelLink, TextSnake, Multi-scale FCN, Holistic-prediction
- end-to-end detectors (including text recognition)
  - Mask TextSpotter
- character-level detectors
  - SegLink, CharGrid
  - WordSup - based on SSD, rect boxes, weakly-supervised, inspiration

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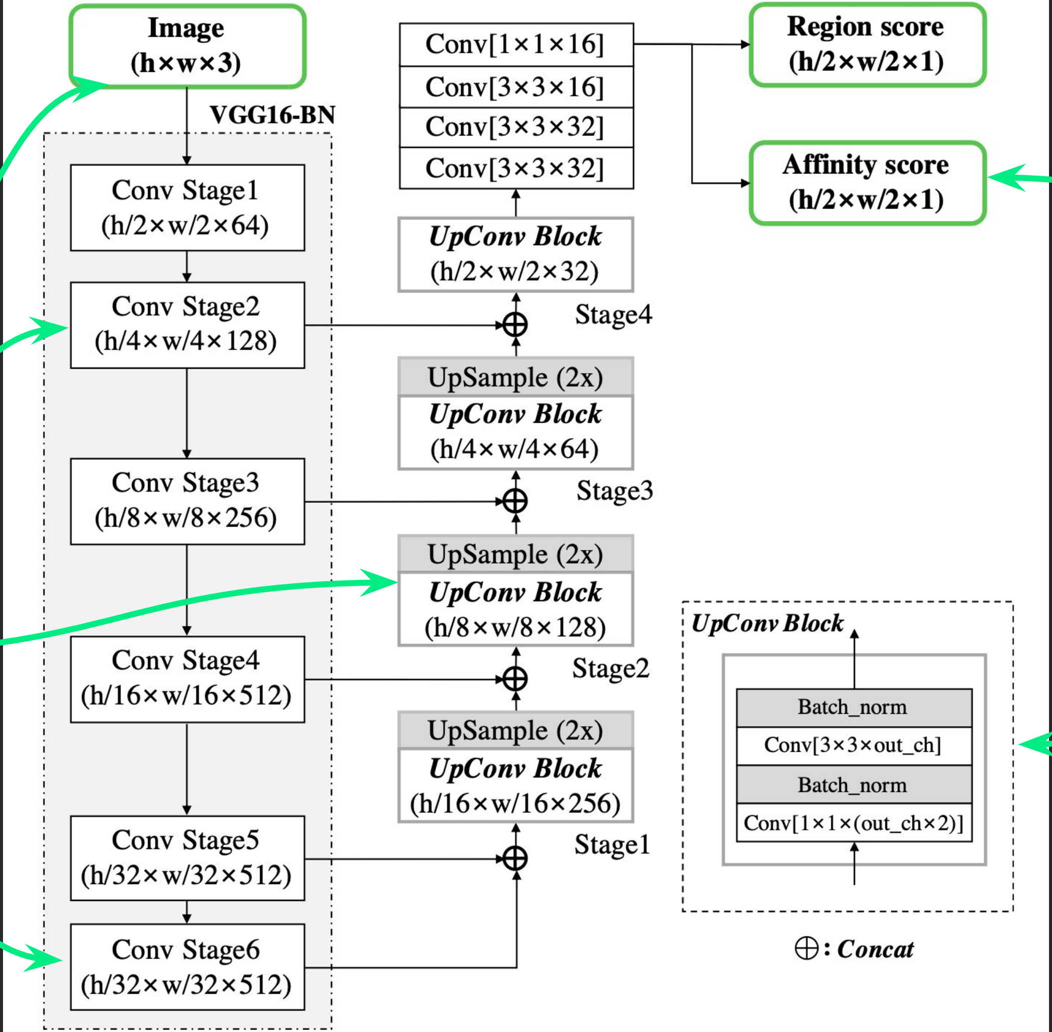
CRAFT architecture

dynamic input shape

VGG-16 backbone

U-net upsampling & skip connections

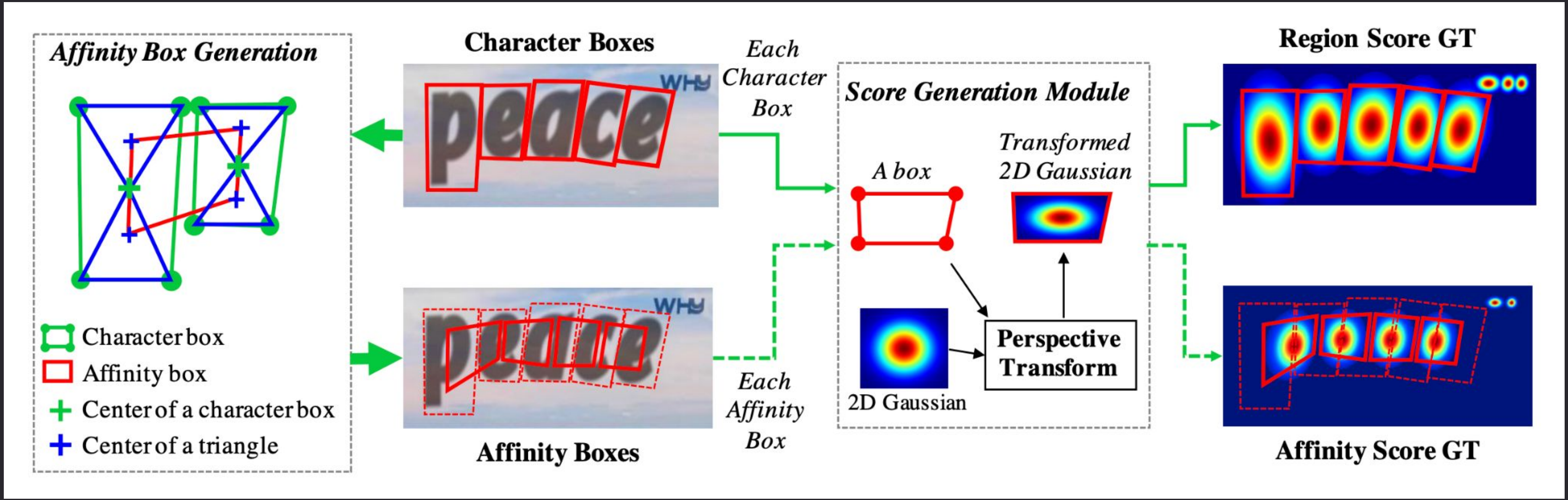
20M params



region & affinity score maps

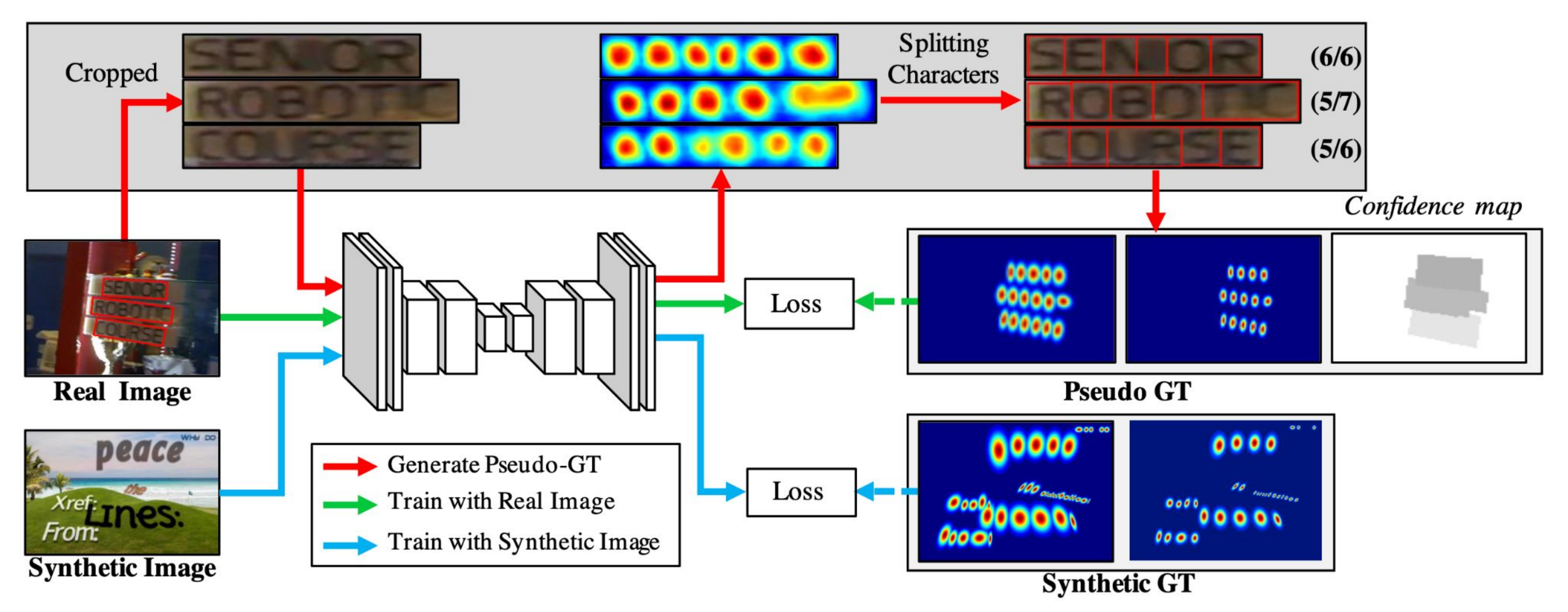
batch normalization

CRAFT – ground-truth labels

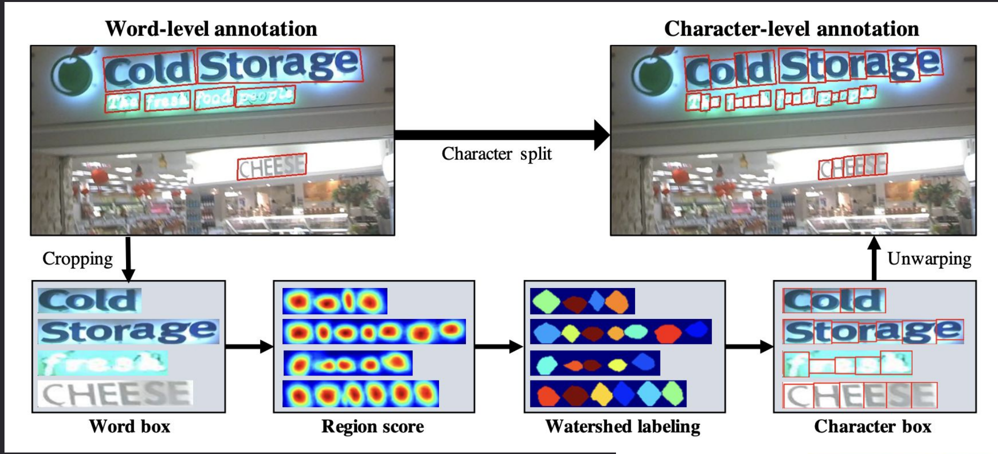


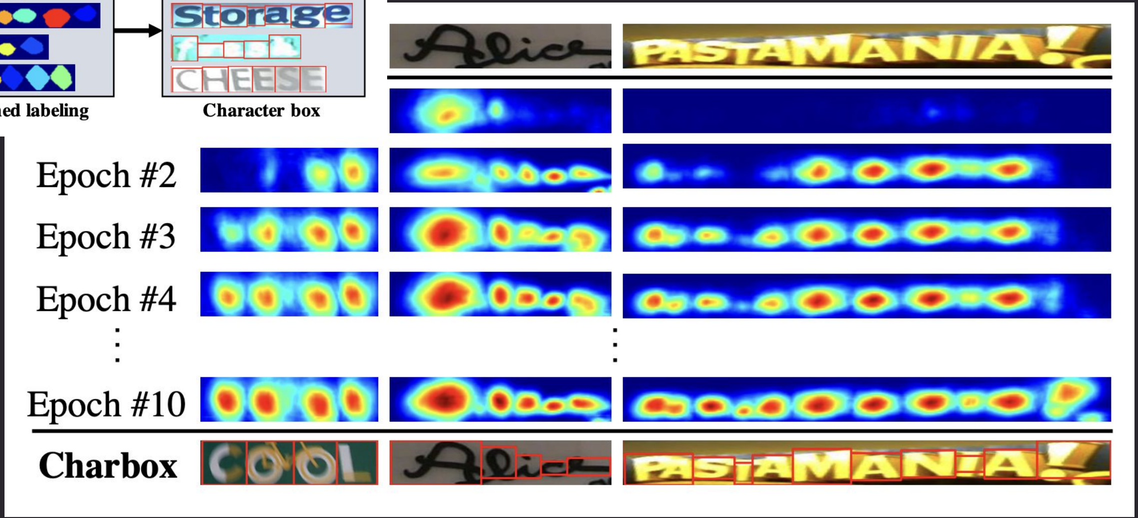
CRAFT – weakly supervised training

synthetic pre-training → fine-tuning with weak labels



CRAFT – weakly supervised training





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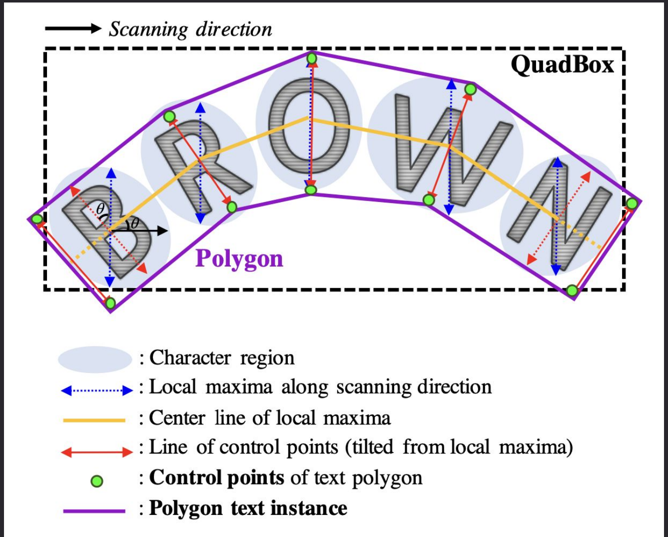
Inference

- word-level oriented rect
  - region & affinity score maps → thresholding → union
  - connected components
  - minimum oriented bounding rect
- char-level rects
  - watershed segmentation
- word-level polygon
  - local maxima → center points
  - distribue max height lines
  - polygon vertices



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Datasets

	shape	text	train size	test size	languages
SynthText	quad	yes	800000	?	multiple
IC13	rectangle	yes	229	233	1 (EN)
IC15	quad	yes	1000	500	1 (EN)
IC17	quad	yes	7200	9000+1800	9
MSRA-TD500	oriented rect	no	300	200	2 (EN+CN)
TotalText	polygon	yes	1255	300	
CTW-1500	polygon	yes	1000	500	1-2 (EN,CN)

Experiments

quadrilateral

Method	IC13(DetEval)			IC15			IC17			MSRA-TD500			FPS
	R	P	H	R	P	H	R	P	H	R	P	H	
Zhang et al. [39]	78	88	83	43	71	54	-	-	-	67	83	74	0.48
Yao et al. [37]	80.2	88.8	84.3	58.7	72.3	64.8	-	-	-	75.3	76.5	75.9	1.61
SegLink [32]	83.0	87.7	85.3	76.8	73.1	75.0	-	-	-	70	86	77	20.6
SSTD [8]	86	89	88	73	80	77	-	-	-	-	-	-	7.7
Wordsup [12]	87.5	93.3	90.3	77.0	79.3	78.2	-	-	-	-	-	-	1.9
EAST* [40]	-	-	-	78.3	83.3	80.7	-	-	-	67.4	87.3	76.1	13.2
He et al. [11]	81	92	86	80	82	81	-	-	-	70	77	74	1.1
R2CNN [13]	82.6	93.6	87.7	79.7	85.6	82.5	-	-	-	-	-	-	0.4
TextSnake [24]	-	-	-	80.4	84.9	82.6	-	-	-	73.9	83.2	78.3	1.1
TextBoxes++* [17]	86	92	89	78.5	87.8	82.9	-	-	-	-	-	-	2.3
EAA [10]	*87*	*88*	*88*	*83*	*84*	*83*	-	-	-	-	-	-	-
Mask TextSpotter [25]	*88.1*	*94.1*	*91.0*	*81.2*	*85.8*	*83.4*	-	-	-	-	-	-	4.8
PixelLink* [4]	87.5	88.6	88.1	82.0	85.5	83.7	-	-	-	73.2	83.0	77.8	3.0
RRD* [19]	86	92	89	80.0	88.0	83.8	-	-	-	73	87	79	10
Lyu et al.* [26]	84.4	92.0	88.0	79.7	89.5	84.3	**70.6**	74.3	72.4	76.2	87.6	81.5	5.7
FOTS [21]	-	-	*87.3*	*82.0*	*88.8*	*85.3*	*57.5*	*79.5*	*66.7*	-	-	-	23.9
CRAFT(ours)	**93.1**	**97.4**	**95.2**	**84.3**	**89.8**	**86.9**	68.2	**80.6**	**73.9**	**78.2**	**88.2**	**82.9**	8.6

polygon

Method	TotalText			CTW-1500		
	R	P	H	R	P	H
CTD+TLOC [38]	-	-	-	69.8	77.4	73.4
MaskSpotter [25]	55.0	69.0	61.3	-	-	-
TextSnake [24]	74.5	82.7	78.4	**85.3**	67.9	75.6
CRAFT(ours)	**79.9**	**87.6**	**83.6**	81.1	**86.0**	**83.5**

end-to-end

Method	IC13	IC15	IC17
Mask TextSpotter [25]	91.7	86.0	-
EAA [10]	90	87	-
FOTS [21]	92.8	**89.8**	70.8
CRAFT(ours)	**95.2**	86.9	**73.9**

Discussion

- 🔎 robustness to scale variance, robust to moire
- 🎓 good generalization ability
- various scripts (at once):
    - ✅ separated chars - OK: Latin, Chinese (中文), Japanese (日本語), Korean (한국어)
    - 😥 cursive/ligatures - char segmentation difficult: Arabic (اَلْعَرَبِيَّةُ), Bengali(বাংলা লিপি)
- detected quads/polys can be warped for better text recognition
- rather simple method, a lot can be done in post-processing
- potential improvements:
    - 🔁 old VGG/U-NET backbone (2014-2015) → modern and efficient convnet
    - joint training with text recognition
    - multi-resolution inference
    - orient char boxes more consistently
- overkill for straight text, good for photos with perspective / crumbled paper

Practical experience

- Pytorch:
  - github.com/clovaai/CRAFT-pytorch + pre-trained Latin model
  - github.com/fcakyon/craft-text-detector
  - `pip install craft_text_detector`
- Keras: github.com/notAI-tech/keras-craft

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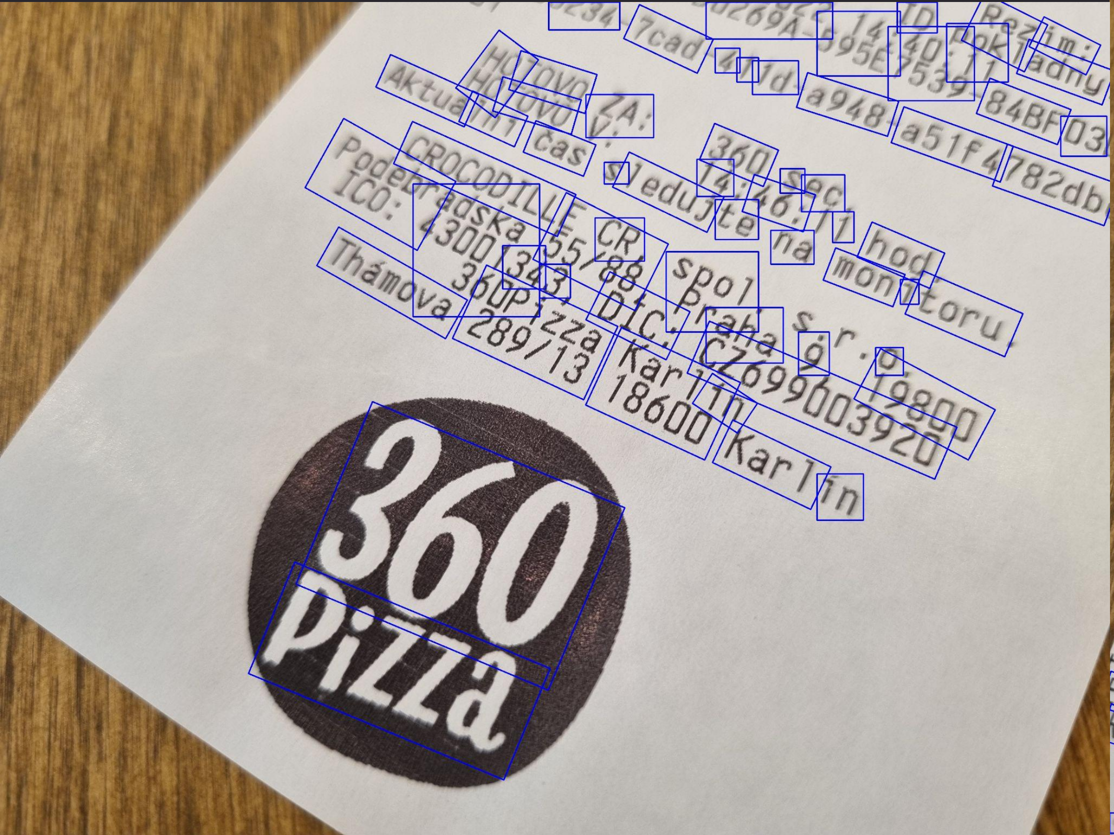
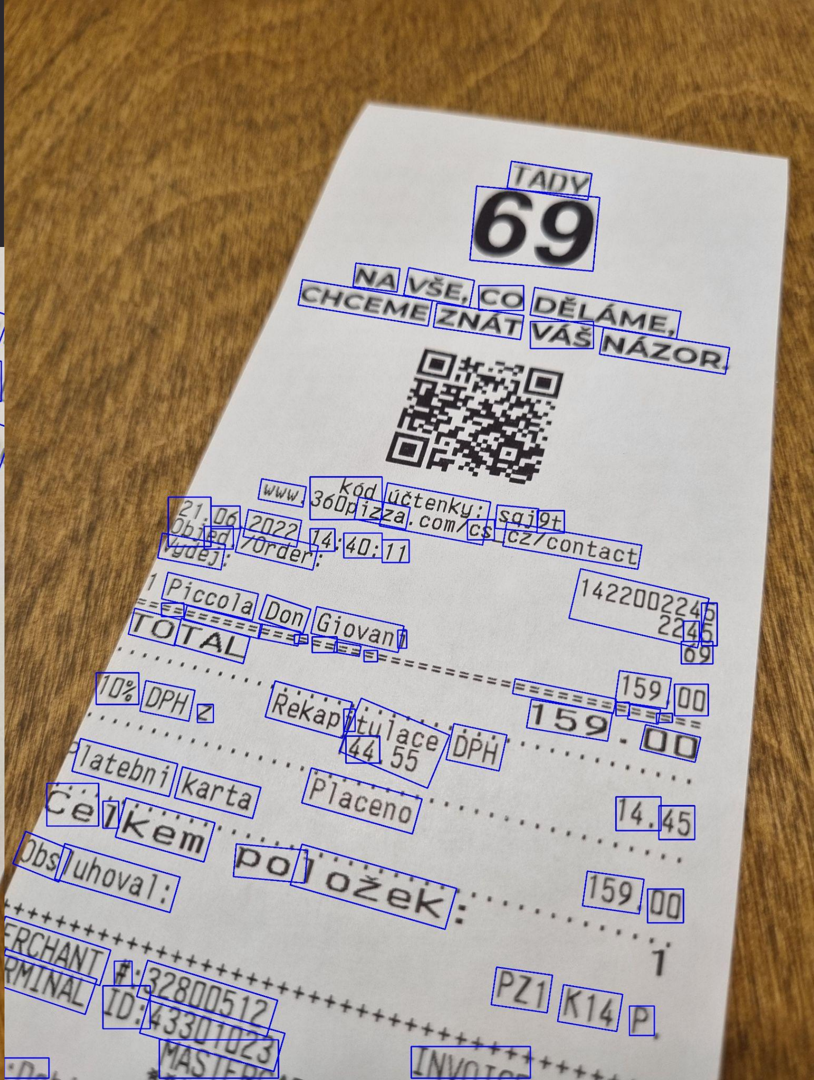
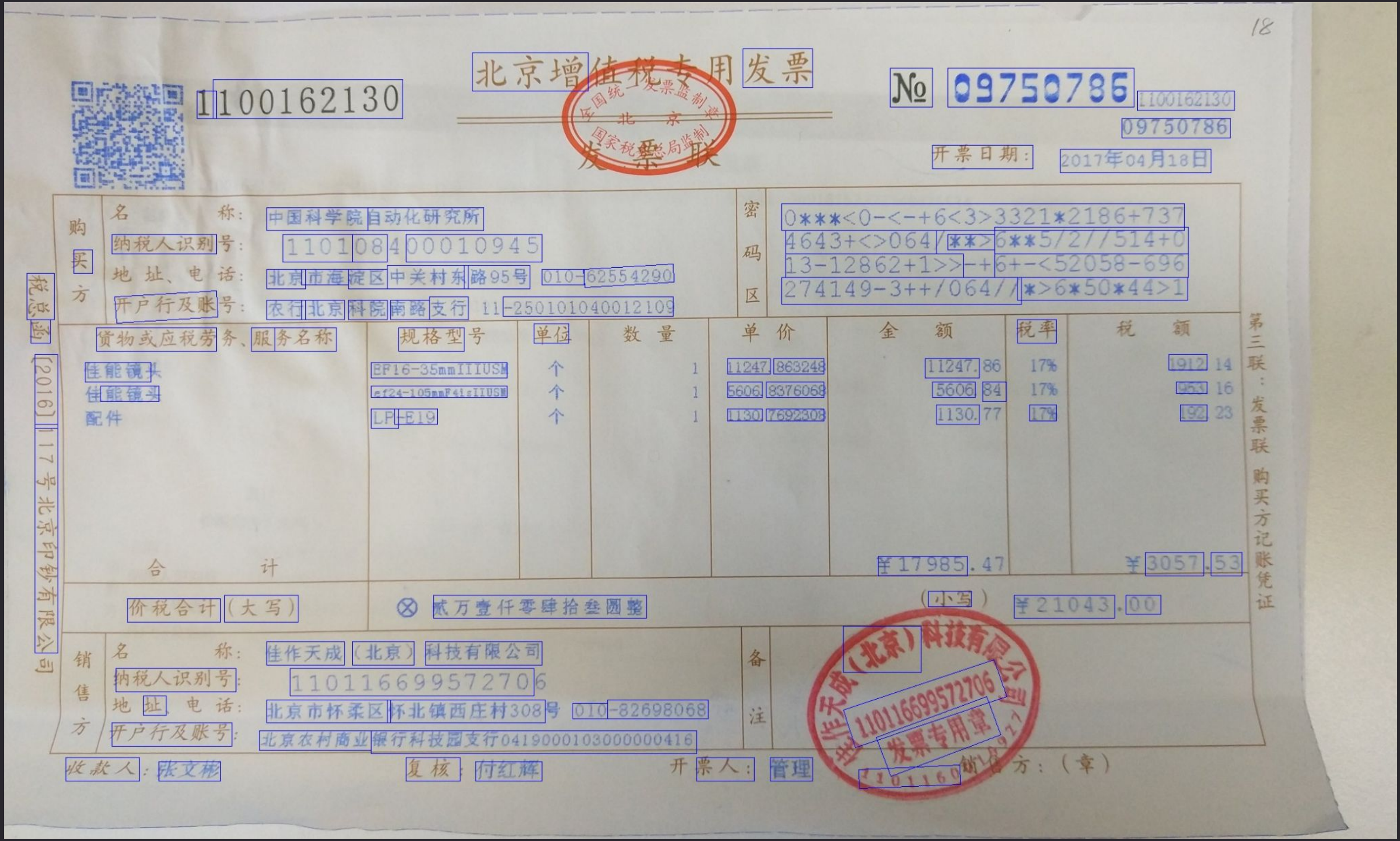
Practical experience

1024x768px → 180 ms (M1 Pro)







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https://github.com/FuxiJia/InvoiceDatasets/blob/master/dataset/images/vat_train/vat_0011.jpg

Thank you

For your attention

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